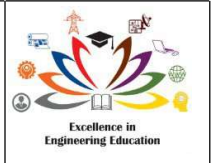




Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY
AND MANAGEMENT, PUNE-09



Pune Vidyarthi Griha's College of Engineering, Technology and Management

Department of Artificial Intelligence and Data Science

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417532(B): Business Intelligence

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Date of Submission: 20-04-2026

ASSIGNMENT 2

AIM:

To extract data from raw sources, transform it into a clean and structured format, and load it into a target system for visualization and analytical insights.

PROBLEM STATEMENT:

Data Visualization from Extract, Transform, Loading process.

THEORY:

In the modern era of data-driven decision-making, Data Visualization is a powerful technique used to represent complex data in an intuitive and interactive manner. However, before data can be visualized effectively, it must go through a series of preparatory steps known as the ETL process — Extraction, Transformation, and Loading. This process is essential in ensuring that the data being visualized is clean, consistent, and meaningful.

1. Extraction

Extraction is the first phase of the ETL process, where raw data is collected from multiple and often heterogeneous sources. These sources can include:

- Flat files like Excel and CSV,
- Relational databases such as SQL Server or Oracle,
- Cloud platforms,
- APIs and web services.

The main goal of extraction is to gather all relevant data without affecting the performance of the source systems. This step ensures that no valuable information is missed during the data preparation process.

2. Transformation

Once data is extracted, it usually exists in different formats, structures, or quality levels. In the transformation phase, the data is cleaned, enriched, standardized, and restructured. Common transformation tasks include:

- Removing duplicates or null values,
- Changing data types (e.g., text to numbers),
- Normalizing data formats (e.g., dates and currencies),
- Creating calculated columns or aggregating data.

Transformation ensures that the data becomes uniform and suitable for accurate analysis and reporting.

3. Loading

The final step in the ETL process is loading the transformed data into a target system or data warehouse, where it becomes ready for visualization. Tools like Power BI or Tableau are then used to connect to this prepared data and create dashboards, charts, and other interactive visual reports.

4. Data Visualization

Once the ETL process is complete, data visualization tools allow users to explore and analyze the data graphically. Visualizations help in:

- Identifying trends and patterns,
- Comparing values across categories,
- Tracking KPIs (Key Performance Indicators),
- Making predictions and business decisions.

Tools like Power BI not only support visualizing data from local files (e.g., Excel) but also allow connections to live databases like SQL Server or cloud services, offering real-time insights.

PROCEDURE :

- **Import Required Libraries:**
The necessary Python libraries such as pandas for data handling and matplotlib for visualization are imported to begin the ETL process.
- **Extract Data from Source:**
An Excel file containing e-commerce sales data is used as the source. The file is opened and the relevant sheet is selected for further processing.
- **Convert Date Column:**
The OrderDate column is converted to a proper datetime format to ensure consistency in date values, which is essential for time-based analysis.
- **Check for Missing Values (Before Cleaning):**
The dataset is examined to identify missing values in each column. The total number of rows before transformation is also recorded to track changes.
- **Transform - Clean the Data:**
Rows containing any missing values are removed to clean the dataset. This step ensures the quality and integrity of the data used for analysis.

- **Recheck After Cleaning:**

The dataset is re-evaluated to verify that all missing values have been removed. The number of records after cleaning is noted for comparison.

- **Load - Save the Cleaned Data:**

The cleaned dataset is exported and saved as a CSV file. This marks the completion of the "Load" phase in the ETL process.

- **Visualize the ETL Record Flow:**

A bar chart is created to visually represent the number of records at each stage—extracted, after transformation, and loaded. This helps in understanding the impact of the transformation process and data cleaning on the dataset size.

CONCLUSION :

The ETL process plays a crucial role in preparing data for meaningful analysis and visualization. Through this assignment, we successfully extracted data from an Excel source, transformed it by handling missing values and ensuring consistency, and then loaded the cleaned data for further use. The final visualization of the ETL stages provided a clear understanding of how data volume and quality are affected during the transformation process. This structured approach not only improves data reliability but also enhances the effectiveness of data visualization, ultimately enabling better business insights and informed decision-making.

```
In [18]: import pandas as pd
import matplotlib.pyplot as plt
```

Load Data

```
In [19]: file_path = "Ecommerce+Sales+Data.xlsx"
excel_file = pd.ExcelFile(file_path)
```

Extract

```
In [20]: sheet_name = excel_file.sheet_names[0]
df = excel_file.parse(sheet_name)
```

Transform

```
In [21]: df['OrderDate'] = pd.to_datetime(df['OrderDate'], errors='coerce')
```

```
In [22]: # Check missing values before transformation
nulls_before = df.isnull().sum()
rows_before = len(df)
nulls_before
```

```
Out[22]: OrderDate          1
OrderNumber         1
ProductKey          1
SalespersonKey      1
Salesperson         1
Supervisor          1
Manager            1
Channel            1
Quantity           1
UnitPrice          1
Total              3015
Month              1
Year              1
dtype: int64
```

```
In [23]: # Drop rows with any nulls (cleaning)
df_cleaned = df.dropna()
df_cleaned
```

Out[23]:

	OrderDate	OrderNumber	ProductKey	SalespersonKey	Salesperson	Supervisor	Manager
0	2022-01-15	1492755.0	180.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
1	2022-01-20	1492618.0	661.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
2	2022-01-24	1492834.0	278.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
3	2022-01-25	1492801.0	1518.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
4	2022-02-01	1501692.0	661.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
...	...	...	...	...	...	...	...
38567	2020-10-22	2075402.0	157.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
38568	2020-10-23	2148884.0	157.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
38569	2020-10-24	2118590.0	157.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
38570	2020-10-25	2104650.0	157.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro
38571	2020-10-26	2152594.0	157.0	215.0	Carla Ferreira	Diego Araujo	Victor Castro

38572 rows × 13 columns

```
In [24]: nulls_after = df_cleaned.isnull().sum()
rows_after = len(df_cleaned)
rows_after
```

Out[24]: 38572

Load

```
In [25]: df_cleaned.to_csv("Cleaned_Ecommerce_Data.csv", index=False)
```

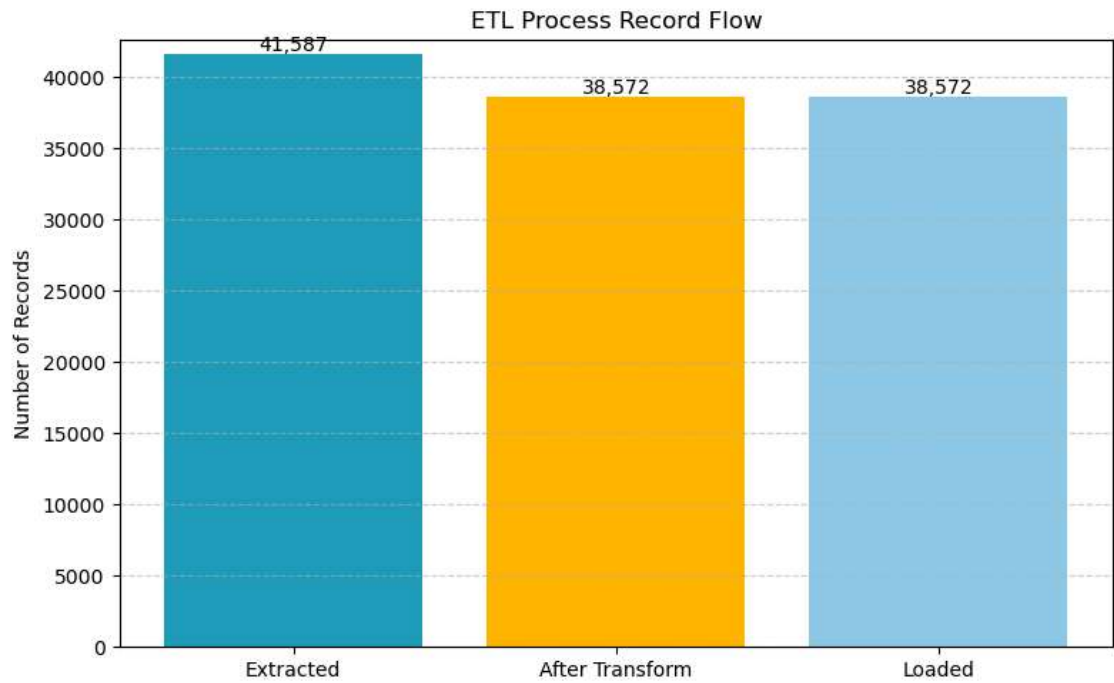
```
In [26]: stages = ['Extracted', 'After Transform', 'Loaded']
record_counts = [rows_before, rows_after, rows_after]
colors = ['#219ebc', '#ffb703', '#8ecae6']
```

Visualization

```
In [27]: plt.figure(figsize=(8, 5))
plt.bar(stages, record_counts, color=colors)
plt.title('ETL Process Record Flow')
plt.ylabel('Number of Records')
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.ylim(0, rows_before + 1000)

for i, count in enumerate(record_counts):
    plt.text(i, count + 200, f"{count:}", ha='center', fontsize=10)
```

```
plt.tight_layout()  
plt.show()
```



In []: